

```

package OnlineMovieBookingApp;

public abstract class User {
    private static int idCounter = 0;
    private int id;
    private String name;

    public User(String name) {
        idCounter += 1;
        this.id = idCounter;
        this.name = name;
    }

    public String getName() {
        return name;
    }
}

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package OnlineMovieBookingApp;

import java.text.ParseException;
import java.text.SimpleDateFormat;
import java.util.ArrayList;
import java.util.Date;
import java.util.HashMap;

public class BookMyShow {
    ArrayList<Theater> theaters;
    static HashMap<String, ArrayList<Show>> movieMap;

    private void generateMovieMap() {
        for (Theater theater : theaters) {
            ArrayList<Show> showArray = theater.getShows();
            for (Show show : showArray) {
                if (show != null) {
                    if (movieMap.containsKey(show.getMovie().getName())) {
                        movieMap.get(show.getMovie().getName()).add(show);
                    } else {
                        ArrayList<Show> movieShowList = new ArrayList<>();
                        movieShowList.add(show);
                        movieMap.put(show.getMovie().getName(),
                                movieShowList);
                    }
                }
            }
        }
    }

    public BookMyShow(ArrayList<Theater> theaters) {
        this.theaters = theaters;
        this.movieMap = new HashMap<>();
        generateMovieMap();
        // System.out.println(movieMap);
    }

    public static ArrayList<Show> searchShows(String movieName) {
        if (movieMap.containsKey(movieName)) {

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        return movieMap.get(movieName);
    } else
        return null;
}

public static void main(String[] args) {
    // Creating Guest User --> Piyush
    GuestUser piyush = new GuestUser("Piyush");
    RegisteredUser ayush = new RegisteredUser("Ayush");
    RegisteredUser saurabh = new RegisteredUser("Saurabh");
    System.out.println(piyush.getName());
    System.out.println(ayush.getName());
    System.out.println(saurabh.getName());

    // Creating Movie object --> Iron Man
    Movie ironMan = new Movie("Iron Man", Language.ENGLISH, Genre.ACTION);
    Movie avengers = new Movie("Avengers: End Game", Language.ENGLISH,
Genre.ACTION);
    Movie walkToRemember = new Movie("The Walk To Remember",
Language.ENGLISH, Genre.ROMANCE);
    Movie housefull = new Movie("HouseFull 2", Language.HINDI, Genre.COMEDY);
    System.out.println(ironMan);
    System.out.println(avengers);
    System.out.println(walkToRemember);
    System.out.println(housefull);

    // Creating Theater
    Theater pvr_gip = new Theater("PVR", "GIP Noida", 30);
    Theater big_cinema = new Theater("Big Cinema", "Sector 137 Noida", 40);
    System.out.println(pvr_gip);
    System.out.println(big_cinema);

    Show show1 = null, show2 = null, show3 = null, show4 = null;
    SimpleDateFormat formatter = new SimpleDateFormat("EEEE, MMM dd, yyyy
HH:mm:ss a");

    try {

        String dateInString = "Friday, Jun 7, 2024 09:00:00 AM";
        Date date = formatter.parse(dateInString);
        show1 = new Show(date, ironMan, pvr_gip);
        System.out.println(show1);

        dateInString = "Friday, Jun 7, 2024 12:00:00 PM";
        date = formatter.parse(dateInString);
        show2 = new Show(date, housefull, pvr_gip);
        System.out.println(show2);

        // BIG-CINEMA
        dateInString = "Friday, Jun 7, 2024 09:00:00 AM";
        date = formatter.parse(dateInString);
        show3 = new Show(date, walkToRemember, big_cinema);
        System.out.println(show3);

        // BIG-CINEMA
        dateInString = "Friday, Jun 7, 2024 12:00:00 PM";
        date = formatter.parse(dateInString);
        show4 = new Show(date, walkToRemember, big_cinema);
        System.out.println(show4);
    } catch (ParseException e) {

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        e.printStackTrace();
    }

    ArrayList<Theater> theaterArrayList = new ArrayList<>();
    theaterArrayList.add(pvr_gip);
    theaterArrayList.add(big_cinema);
    System.out.println("Theatre list : " + theaterArrayList);

    BookMyShow bookMyShow = new BookMyShow(theaterArrayList);
    ArrayList<Show> searchedShow = BookMyShow.searchShows("Iron Man");
    Show bookingShow = searchedShow.get(0);
    System.out.println("bookingShow : " + bookingShow);

    TicketBookingThread t1 = new TicketBookingThread(bookingShow, ayush, 10);
    TicketBookingThread t2 = new TicketBookingThread(bookingShow, saurabh, 10);
    TicketBookingThread t3 = new TicketBookingThread(bookingShow, saurabh, 10);
    t1.start();
    t2.start();
    t3.start();

    try {
        t1.join();
        t2.join();
        t3.join();
    } catch (InterruptedException e) {
        e.printStackTrace();
    }

    Ticket ayush_ticket = t1.getTicket();
    Ticket saurabh_ticket = t2.getTicket();
    Ticket saurabh_ticket2 = t3.getTicket();
    System.out.println(ayush_ticket);
    System.out.println(saurabh_ticket);
    System.out.println(saurabh_ticket2);

    searchedShow = BookMyShow.searchShows("HouseFull 2");
    bookingShow = searchedShow.get(0);

    TicketBookingThread t4 = new TicketBookingThread(bookingShow, ayush, 15);
    TicketBookingThread t5 = new TicketBookingThread(bookingShow, saurabh, 10);
    // t5.start();
    t4.start();
    try {
        t4.join();
        t5.join();
    } catch (InterruptedException e) {
        e.printStackTrace();
    }
    Ticket ayushNewTicket = t4.getTicket();
    // Ticket saurabhNewTicket = t5.getTicket();
    System.out.println(ayushNewTicket);
    // System.out.println(saurabhNewTicket);
}
}

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package OnlineMovieBookingApp;

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public enum Genre {  
    ACTION, ROMANCE, COMEDY, THRILLER, HORROR  
}
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```
package OnlineMovieBookingApp;  
  
public class GuestUser extends User {  
    public GuestUser(String name) {  
        super(name);  
    }  
  
    public void register(String username, String  
password) {  
    }  
}
```

```
package OnlineMovieBookingApp;  
  
public enum Language {  
    TELUGU, TAMIL, ENGLISH, HINDI  
}
```

```
package OnlineMovieBookingApp;  
  
public class Movie {  
    private String name;  
    private float rating = 0.0f;  
    private Language language;  
    private Genre genre;  
  
    public Movie(String name, Language language, Genre  
genre) {  
  
        this.name = name;  
        this.language = language;  
        this.genre = genre;  
    }  
  
    public String getName() {
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        return name;
    }

    public float getRating() {
        return rating;
    }

    public Language getLanguage() {
        return language;
    }

    public Genre getGenre() {
        return genre;
    }

    @Override
    public String toString() {
        return "Movie [name=" + name + ", language=" +
language + ", genre=" + genre + "]\n";
    }
}

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package OnlineMovieBookingApp;

import java.util.ArrayList;

public class RegisteredUser extends User {
    public ArrayList<Ticket> bookingHistory;

    public RegisteredUser(String name) {
        super(name);
        this.bookingHistory = new ArrayList<>();
    }

    public void cancelTicket(Ticket ticket) {
    }
}

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```

package OnlineMovieBookingApp;

import java.util.Date;

```

```

public class Show {
    private static int idCounter = 0;
    private int id;
    private Date showTime;
    private Movie movie;
    private Theater theater;
    private int availableSeats;

    public Show(Date showTime, Movie movie, Theater
theater) {
        idCounter += 1;
        this.id = idCounter;
        this.showTime = showTime;
        this.movie = movie;
        this.theater = theater;
        this.availableSeats = theater.getCapacity();
        theater.getShows().add(this);
    }

    public Movie getMovie() {
        return movie;
    }

    public void setTheater(Theater theater) {
        this.theater = theater;
    }

    public void setAvailableSeats(int availableSeats) {
        this.availableSeats = availableSeats;
    }

    public int getAvailableSeats() {
        return availableSeats;
    }

    public void updateShow() {
    }

    public synchronized Ticket bookTicket(RegisteredUser
user, int seats) {
        if (availableSeats >= seats && seats > 0) {
            Ticket ticket = new Ticket();
            availableSeats -= seats;
            ticket.setOwner(user.getName());
            ticket.setBookedShow(this);
        }
    }
}

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        ticket.setBookingTime(new Date());
        ticket.setNumberOfSeats(seats);
        System.out.println("Successfully booked");
        user.bookingHistory.add(ticket);
        return ticket;
    } else {
        System.out.println("Seats not Available");
        return null;
    }
}

@Override
public String toString() {
    return "Show{" + "id=" + id + ", showTime=" +
showTime + ", movie=" + movie.getName() + ", theater="
        + theater.getName() + ",
availableSeats=" + availableSeats + '}';
}
}

```

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package OnlineMovieBookingApp;

```

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import java.util.ArrayList;

```

```

public class Theater {
    private static int idCounter = 0;
    private int id;
    private String name;
    private String location;
    private int capacity;
    private ArrayList<Show> shows;

    public Theater(String name, String location, int
capacity) {
        idCounter += 1;
        this.id = idCounter;
        this.name = name;
        this.location = location;
        this.capacity = capacity;
        this.shows = new ArrayList<>();
    }

    public void updateShow(Show oldShow, Show newShow) {
    }
}

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```

    public ArrayList<Show> getShows() {
        return shows;
    }

    public String getName() {
        return name;
    }

    public int getCapacity() {
        return capacity;
    }

    @Override
    public String toString() {
        return "Theater [id=" + id + ", name=" + name +
", location=" + location + ", capacity=" + capacity + ",
shows="
        + shows + "]\n";
    }
}

```

```

package OnlineMovieBookingApp;

import java.util.Date;

public class Ticket {
    private static int idCounter = 0;
    private int id;
    private String owner;
    private Date bookingTime;
    private int numberOfSeats;
    private Show bookedShow;

    public Ticket() {
        idCounter += 1;
        this.id = idCounter;
    }

    public String getTicketInfo() {
        return null;
    }
}

```



```
public int cancelTicket() {  
    return 0;  
}  
  
public String getOwner() {  
    return owner;  
}  
  
public void setOwner(String owner) {  
    this.owner = owner;  
}  
  
public int getId() {  
    return id;  
}  
  
public void setId(int id) {  
    this.id = id;  
}  
  
public Date getBookingTime() {  
    return bookingTime;  
}  
  
public void setBookingTime(Date bookingTime) {  
    this.bookingTime = bookingTime;  
}  
  
public int getNumberOfSeats() {  
    return numberOfSeats;  
}  
  
public void setNumberOfSeats(int numberOfSeats) {  
    this.numberOfSeats = numberOfSeats;  
}  
  
public Show getBookedShow() {  
    return bookedShow;  
}  
  
public void setBookedShow(Show bookedShow) {  
    this.bookedShow = bookedShow;  
}
```

```

        @Override
        public String toString() {
            return "Ticket [id=" + id + ", owner=" + owner +
", bookingTime=" + bookingTime + ", numberOfSeats="
+ numberOfSeats + ", bookedShow=" +
bookedShow + "]\n";
        }
    }
}

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package OnlineMovieBookingApp;

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```

public class TicketBookingThread extends Thread {
    private Show show;
    private RegisteredUser user;
    private int numberOfSeats;
    private Ticket ticket;

    public TicketBookingThread(Show show, RegisteredUser
user, int numberOfSeats) {
        this.show = show;
        this.user = user;
        this.numberOfSeats = numberOfSeats;
    }

    @Override
    public void run() {
        ticket = show.bookTicket(user, numberOfSeats);
    }

    public Ticket getTicket() {
        return ticket;
    }
}

```